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In class, we discussed fitting human blood viscosity data using the Sisko Model for a shear-rate-dependent viscosity $\eta(\dot{\gamma})$:
 $\eta - \eta_{\infty} = m \dot{\gamma}^{n-1}$ $\eta = m \dot{\gamma}^{n-1} + \eta_{\infty}$ $y = ax + b$ $x = \sqrt[n]{\frac{y-b}{a}}$
where η_{∞} , m , and n are three adjustable constants. It is known that for human blood that $n = 0.5$ is a good approximation.

- (a) Assuming that $n = 0.5$, how can the Sisko model be linearized so that linear least squares can be used to determine the remaining two constants? Show clearly what you would plot as x on the horizontal axis and y on the vertical axis.
- (b) Use the data provided to calculate the remaining two parameters using horizontal offsets. Be sure to give the correct units.

For horizontal offsets: slope $a = \frac{S_{yy} - S_y^2}{S_{xy} - S_x S_y}$ and intercept $b = \frac{S_{xy} S_y - S_{yy} S_x}{S_{xy} - S_x S_y}$.

$$d_i = x_i - x(\bar{y}_i)$$

Shear Rate $\dot{\gamma}$ (s^{-1})	Viscosity η (mPa·s)
0.51	39.15
1.34	22.18
7.97	9.51
90.73	5.77
759.07	4.23

(A) In order to linearize, also follow $y = ax + b$ format. You need the y -intercept (η_{∞}) & m :

• $n = 0.5$ $\eta = m \dot{\gamma}^{(0.5)-1} + \eta_{\infty} \rightarrow \eta = m \dot{\gamma}^{-0.5} + \eta_{\infty}$

• plug in a known $\dot{\gamma}$ & η value.
Assuming $\eta_{\infty} = 0$.

$$(22.18 \text{ mPa}\cdot\text{s}) = m (1.34)^{-0.5} + (0)$$

$$(22.18 \text{ mPa}\cdot\text{s}) = m (1.34)^{-0.5}$$

$$m = 25.675 \text{ mPa}\cdot\text{s}^2$$

* $\dot{\gamma}^{-0.5}$ on the x -axis
* η on the y -axis
yes!

Solve for η_{∞} : $(9.51 \text{ mPa}\cdot\text{s}) = (25.675 \text{ mPa}\cdot\text{s}^2) (7.97)^{-0.5} + \eta_{\infty}$

$$(9.51 \text{ mPa}\cdot\text{s}) = (9.09 \text{ mPa}\cdot\text{s}) + \eta_{\infty}$$

$$\eta_{\infty} = 0.41537$$

$$\eta = (25.675 \text{ mPa}\cdot\text{s}^2) \dot{\gamma}^{-0.5} + (0.41537)$$

$S_y =$

$S_x =$

(B) $d_i = x_i - x(\bar{y}_i)$ where $\bar{y}_i = \left[\dot{\gamma} = \sqrt[n]{\frac{\eta - \eta_{\infty}}{m}} \right]$:

$$d_i = x_i - x \left[\sqrt[n]{\frac{\eta - \eta_{\infty}}{m}} \right] \rightarrow d_i^2 = \left(x_i - x \left[\sqrt[n]{\frac{\eta - \eta_{\infty}}{m}} \right] \right)^2 = \phi(m, \eta_{\infty})$$

$$(a) = \frac{S_{yy} - S_y^2}{S_{xy} - S_x S_y}$$

$$(b) = \frac{S_{xy} S_y - S_{yy} S_x}{S_{xy} - S_x S_y}$$

